Original Article

Formulation and evalution of new polyherbal toothpaste for oral care

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Abstract

Aim and Objective: The main objective of this research was to formulate medicated polyherbal toothpaste with better antimicrobial activity from natural sources for dental infections.

Materials and Method: The hydroalcoholic extracts of apple peel, lemon peel, orange peel, banana peel was carried out and the formulated toothpaste from the above extracts were subjected for antimicrobial activity study against Gram-positive bacteria (*Staphylococcus aureus, Enterococcus faecalis, Streptococcus mutans*) and Gram-negative bacteria (*Porphyromonas gingivalis*).

Results: Antibacterial activity of the toothpaste was determined by the 2-fold serial dilution method. All the extract has showed good antimicrobial against *P. gingivalis*, with apple extract showing the best result (7.8 mg/ml).

Conclusion: Thus, our study shows the potent antibacterial activity of the formulated polyherbal toothpaste compared with the marketed formulation against dental caries.

Key Words: Antimicrobial efficacy, dental infections, polyherbal toothpaste

Introduction

Dental caries and periodontal disease are complex multifactorial diseases with a dental plaque as their primary cause.[1] Caries and periodontal disease can be prevented by the good maintenance of oral hygiene with the use of oral care products such as toothpaste, toothbrush, mouthwash, and toothpaste that contain antimicrobial properties. Nowadays, there are high interest in oral care products that are incorporated with medicinal plant extracts and are used extensively by the consumers due to low toxicity as compared to oral care products containing antimicrobial agents such as triclosan, cetylpyridinium chloride, chlorhexidine, and amine fluorides that are reported to exhibit toxicity and cause staining of the teeth.[2,3] It is well established that many metabolites produced by plant extracts such as tannins, terpenoids, alkaloids, and flavonoids provide a new source of antimicrobial substances that help in combating new developing drug resistant pathogens.[4-6]

In India, it has been found that a significant proportion of dental problems are due to microbial infections. Dental problems are of three types, formation of dental plaques, dental caries, and periodontal diseases. Dental caries results from the accumulation of plaque on the surface of the teeth and biochemical activities of complex micro-organisms. *Streptococcus mutans* is one of the main opportunistic microbes, which play a vital role in fermenting carbohydrates resulting in acid production and leading to the demineralization of the tooth enamel.

Patients prefer to have an attractive smile, as it is considered to be synonymous with good health. This growing demand for an enhanced esthetic appeal has led to the great development of bleaching products.[7] Bleach and peroxide are commonly used as whitening agents in commercial toothpastes. However, both bleach and peroxide can be an irritant to the mouth and skin in small doses, and are considered to be hazardous materials because they can cause severe chemical burns in large doses. Artificial flavorings and scents are commonly made from synthetic chemicals derived from petrochemicals or coal tar. The color of teeth is influenced by a combination

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of their intrinsic color and extrinsic stains that may form on the tooth surface. The major components of the whitening toothpastes include surfactants, polyphosphates, and enzymes. Some of them also contain low peroxide concentrations. However, abrasive agents are the primary stain removal ingredient present in these toothpastes.[8]

India has a rich tradition of indigenous medicines for various treatments. However, nowadays people are more inclines toward the use of allopathic drugs. Hence, the traditional use of Ayurveda has come down a country which has medicine significant back more than a 3000 years and perhaps combinations of herbs must be known since then, but now people have all along been using drugs and the age-old techniques have been forgotten. After studying the drawbacks of commercial toothpastes, people are now more inclined toward the use of nonalcoholic toothpastes and herbal formulations. Herbal toothpaste does not contain the artificial colors, flavors or fluoride that many of the artificial products contain.

Based on the above information an attempt is made to formulate new polyherbal toothpaste and to study its anti-microbial potency.

Materials and Methods

Collection of fruits

The fresh apple fruits, lemon fruit, orange fruit, banana fruit, was collected from Reliance Fresh Belagavi, Karnataka, in the month of January 2014 [Table 1].

Authentication of the fruit was done by Dr. Harsha Hegde, Scientist, Regional Medical Research Center (RMRC), Belagavi, Karnataka.

Preparation of crude extract

The peels of the fruits were taken off using a peeler and were kept for shade drying at room temperature for a week to avoid chemical degradation due to sunlight.

Table 1: Name of the fruits collected with its botanical name and family

Name of extracts	Botanical name	Family
Apple peel extract	*Malus domestica*	Rosacea
Orange peel extract	*Citrus sinensis* (L)	Rutaceae
Lemon peel extract	*Citrus limon*	Rutaceae
Banana peel extract	*Musa acuminata* Colla.	Musaceae
Clove oil	*Syzygium aromaticumb*	Myrtaceae

Demineralized water

Grinding of the dried peels was done, with the aid of a grinder and converted into coarse powder to the size - 0.1 micron. The crude powder was defatted with pet ether (40–60) and then the defatted material was subjected to extraction with hydro alcohol using a soxhlet apparatus. The extract was concentrated with the help of a rotary evaporator. The excess solvent present was collected and reused; the final semisolid extracts were stored at cold temperature (3–5°C) for further studies.[9,10]

Preparation of bacterial suspension

We used three facultative anaerobic Gram-positive bacteria, which included against *Staphylococcus aureus* ATCC No. 12598, *Enterococcus faecalis* ATCC No. 35550, *S. mutans* ATCC No. 2517 Gram-negative periodontopathi bacteria, *Porphyromonas gingivalis* ATCC No. 33277, in the present study. The bacterial stock cultures (stored at −80°C freezer) were obtained and the work was done at Basic Science Research Center (BSRC), KLE University, Belagavi, Karnataka.

The bacterial stock cultures were thawed and placed in brain heart infusion (BHI) broth for aerobic bacteria and incubated at 37°C for 24 h.[11,12] The anaerobic bacteria were cultured in thioglycolate broth and incubated at 37°C, 5% CO_2 for 48 h. The inoculum was prepared by adjusting the turbidity of the suspensions to 0.5 McFarland standards earlier to minimum inhibitory concentration (MIC) testing. The bacteria were then tested using as a standard drug as ciprofloxacin.

Determination of minimum inhibitory concentration

Four extracts (apple peel, banana peel, lemon peel and orange peel) 50% stock solution was prepared that is, (500 mg if the extract in 10% of dimethyl sulfoxide solution).

In the tubes, 200 µl of extract and 200 µl of BHI broth was added the nine tubes separately. From second tube 200 µl was transferred to third tube this serial dilution was repeated up to 10^{-8} dilution for each extract. From the maintained stock cultures of required organisms, 5 µl was taken and added into 2 ml of BHI broth, in each serially diluted tube, 200 µl of above culture suspension was added. The last 10th tube contained only the media and culture suspension, the highest concentration taken was 500 mg/ml and the lowest concentration taken was 1.95 mg/ml. The tubes were kept for incubation for 24 h at 37°C in a bacteriological incubator and observed for turbidity.[12]

The same procedure was also used for the determination of MIC of the marketed formulation and polyherbal toothpaste.

Preparation of the formulation

Based on the sensitivity and the resistance observed by MIC results of the each extracts, the formulation was done.[13] Where 5 g of the apple peel extract, 5 g of orange peel extract, 5 g of banana peel extract, 5 g of lemon peel extract was mixed and triturated with 5 g of carbopol 934 and 1 g of sodium benzoate was added as a preservative in a mortar and pestle. To it 2 ml of Tween 80 was added to remove the stickiness and further 1 ml of glycerin was added triturated well, to the above content 80 ml of demineralized water was added and finally made up to 100 g. pH was adjusted with a solution of sodium hydroxide. Clove oil was added at last to mask the bitter taste and to give anti-inflammatory effect.

Results

Antimicrobial studies of the extracts

The MIC of the hydroalcoholic extracts was specified in Table 2. The low MIC value indicates that the extract has strong antibacterial activity the result revealed the hydroalcoholic extract of the *Malus domestica*, Figure 1 shows the serial dilution method conducted for the antimicrobial study of apple peel ectract (*Malus domestica*) against (*S.mutans*) organism. *Citrus sinensi* (L.), *Citruslimon*, *Musa acuminata* Colla. Possess good antibacterial activity against both Gram-positive and Gram-negative bacteria. The above result helped us to determine in what concentration the extract should be used in the formulation of toothpaste.

Antimicrobial studies of the formulation

The MIC of the formulated toothpaste containing hydroalcoholic extracts were specified in Table 3 the low MIC value indicates that the extract has strong and equivalent antibacterial activity the result revealed the hydroalcoholic extract of the *M. domestica*, *C. sinensi* (L.), *Citruslimon*, *M. acuminata* Colla. Possess good antibacterial activity against both Gram-positive and Gram-negative bacteria.[14,15,16]

Discussion

The extracts studied for antibacterial activity showed good sensitivity toward the bacteria *S. aureus*, *P. gingivalis*, *S. mutans*, *and E. faecalis* for different dilutions. As shown in Table 2, in which all the extracts have shown good sensitivity against all organisms. All the extracts are more sensitive to *P. gingivalis* as compared to other organism. Apple peel 7.8 mg/ml, banana peel 31.25 mg/ml, lemon peel 15.63 mg/ml, orange peel 62.5 mg/ml. The formulated polyherbal toothpaste was

Table 2: MIC results of the different extracts (apple, banana, lemon, orange peels) on different organisms at different concentrations

Extracts/microorganism	*S. mutans*	*S. aureus*	*E. faecalis*	*P. gingivalis*
Ciprofloxacin (µg/ml)	1	2	1	1
Apple peel (mg/ml)	62.5	62.5	15.63	7.8
Banana peel (mg/ml)	500	62.5	500	31.25
Lemon peel (mg/ml)	62.5	31.25	31.25	15.63
Orange peel (mg/ml)	125	62.5	250	62.5

MIC: Minimum inhibitory concentration, *S. mutans: Streptococcus mutans, S. aureus: Staphylococcus aureus, E. faecalis: Enterococcus faecalis, P. gingivalis: Porphyromonas gingivalis*

Table 3: MIC results of polyherbal toothpaste (comparison with market herbal toothpaste)

Sample	100 µg/ml	50 µg/ml	25 µg/ml	12.5 µg/ml	6.25 µg/ml	3.125 µg/ml	1.6 µg/ml	0.8 µg/ml	0.4 µg/ml	0.2 µg/ml
Market herbal toothpaste										
P. gingivalis	S	S	S	S	S	S	S	S	S	S
S. aureus	S	S	S	S	S	S	S	S	S	S
E. faecalis	S	S	S	S	S	S	S	S	S	R
S. mutans	S	S	S	S	S	S	S	S	S	S
Polyherbal tooth paste										
P. gingivalis	S	S	S	S	S	S	S	S	S	S
S. aureus	S	S	S	S	S	S	S	S	S	R
E. faecalis	S	S	S	S	S	S	S	S	S	S
S. mutans	S	S	S	S	S	R	R	R	R	R

S: Sensitive, R: Resistant, *P. gingivalis: Porphyromonas gingivalis, S. aureus: Staphylococcus aureus, E. faecalis: Enterococcus faecalis, S. mutans: Streptococcus mutans,* MIC: Minimum inhibitory concentration



Figure 1: Image of the antibacterial study-serial dilution method

also screened for the antibacterial activity against the four microorganisms (*S. aureus*, *P. gingivalis*, *S. mutans*, *and E. faecalis*). The polyherbal toothpaste showed promising results, similar to the marketed toothpaste, at all the dilutions, against *P. gingivalis* and *E. faecalis* whereas it showed sensitivity up to 0.4 µg/ml against *S. aureus*. The MIC of the test formulation against *S. mutans* was 6.25 µg/ml, which was very high when compared to the marketed toothpaste. Demineralized water has been used in this formulation, instead of alcohol; hence this formulation is free from alcohol and not habit forming. Glycerin has been added which is an astringent and also helps in strengthening the gums of the oral cavity. Hence, the polyherbal toothpaste formulated is of much advantage when compared to the marketed toothpaste.

Conclusion

A study effort has been made for preparing new polyherbal toothpaste by combining the different fruit peel extracts against organisms causing dental caries and infections. For the present study apple, lemon, banana, orange were subjected for extraction and these extracts were subjected for the anti-microbial activity. Finally, the polyherbal toothpaste was tested for antibacterial activity and was compared with commercial toothpaste. The result shows significant antibacterial potency for the polyherbal toothpaste for commercial use.

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